

| Clusters   | Tasks        | Metrics | GPT-3.5      | GPT-4        | LLaMA2 | Vicuna | InstructGraph-INS |
|------------|--------------|---------|--------------|--------------|--------|--------|-------------------|
| Structure  | Conn. Dect.  | ACC     | 81.45        | 80.47        | 54.01  | 54.85  | <b>83.54</b>      |
|            | Cycle Dect.  | ACC     | 59.02        | 61.44        | 50.79  | 52.88  | <b>91.10</b>      |
|            | Hami. Path   | ACC     | 21.03        | 29.10        | 1.23   | 1.23   | <b>34.80</b>      |
|            | Bipt. Match  | ACC     | 50.23        | 66.11        | 0.00   | 0.00   | <b>76.36</b>      |
|            | Shrt. Path   | ACC     | 38.99        | 49.03        | 0.00   | 0.00   | <b>66.29</b>      |
|            | Degree Comp. | ACC     | 41.18        | <b>70.59</b> | 18.13  | 19.57  | 65.65             |
| Caption    | Wikipedia    | BLEU    | 91.99        | 93.85        | 77.15  | 82.94  | <b>95.81</b>      |
|            | WebNLG       | BLEU    | <b>99.51</b> | 99.29        | 88.67  | 89.33  | 97.35             |
|            | GenWiki      | BLEU    | 98.60        | <b>98.65</b> | 79.72  | 87.67  | 97.71             |
|            | EventNA      | BLEU    | 62.66        | 61.75        | 53.39  | 75.52  | <b>81.64</b>      |
|            | Xalign       | BLEU    | 86.77        | 88.59        | 84.05  | 86.05  | <b>93.08</b>      |
| Graph QA   | PathQSP      | EM      | 52.54        | 68.64        | 42.70  | 31.90  | <b>86.40</b>      |
|            | GrailQA      | EM      | 43.92        | 60.17        | 15.83  | 17.95  | <b>81.30</b>      |
|            | WebQSP       | EM      | 53.73        | 61.57        | 40.07  | 26.42  | <b>73.30</b>      |
|            | WikiTQ       | EM      | 49.02        | <b>60.78</b> | 29.94  | 35.76  | 47.82             |
| Node CLS   | Cora         | EM      | 74.51        | 64.17        | 83.04  | 84.08  | <b>89.33</b>      |
|            | Citeseer     | EM      | 70.39        | <b>74.94</b> | 68.24  | 67.94  | 71.65             |
|            | Pubmed       | EM      | 74.63        | 77.16        | 79.78  | 80.18  | <b>81.09</b>      |
|            | Arxiv        | EM      | 70.59        | 74.51        | 45.50  | 57.75  | <b>81.50</b>      |
|            | Products     | EM      | 68.82        | 84.16        | 29.34  | 79.50  | <b>95.20</b>      |
| Link Pred. | Wikidata     | Hits@1  | 43.73        | 62.94        | 10.75  | 10.38  | <b>96.52</b>      |
|            | FB15K-237    | Hits@1  | 60.34        | 66.88        | 0.00   | 0.00   | <b>98.91</b>      |
|            | ConceptNet   | Hits@1  | 31.33        | 38.30        | 8.30   | 8.19   | <b>59.86</b>      |
| Relevance  | Wikipedia    | ACC     | 94.40        | <b>100</b>   | 69.27  | 68.12  | <b>100</b>        |
| RecSys     | Amazon       | Hits@1  | 27.09        | 59.77        | 44.40  | 16.40  | <b>78.80</b>      |
| IE         | Wikipedia    | F1      | 50.97        | 46.89        | 40.76  | 38.84  | <b>83.56</b>      |
|            | UIE          | F1      | 24.41        | 26.22        | 20.21  | 26.11  | <b>76.82</b>      |
|            | InstructKGC  | F1      | 21.44        | 21.86        | 19.26  | 16.6   | <b>38.98</b>      |
| Graph Gen. | NLGraph      | F1      | 80.86        | 88.17        | 3.64   | 42.21  | <b>91.05</b>      |
| Avg.       |              |         | 59.45        | 66.76        | 41.65  | 46.06  | <b>79.84</b>      |